

Improved Extended Linear Mixing Model in Blind Hyperspectral Unmixing to Address Flexible Endmember Variability

Afshin Ahmadloo, Sadegh Samadi

Abstract—Spectral unmixing is a key aspect in hyperspectral imaging that deals with separating different sources to recognize and categorize various materials at different wavelengths, known as endmember signatures. Abundance signifies the ratio of these EMs. The extraction method of the aforementioned information is not uniform due to the variability of endmembers and nonlinear mixing phenomena. In 2016, Drumetz proposed the Extended Linear Mixing Model (ELMM) to address endmember variability during the unmixing process. Although this strategy yields favorable results in numerous cases, certain limitations in its application have arisen. The model exhibits intrinsic ambiguity when endmembers remain constant, although variations in material structure (microscopic effects) are present. Furthermore, the issue of source separation persists in tiny nonlinear mixtures within composite materials. So, a deeper theoretical insight into the formulation is required. This research proposes a novel model, termed "improved ELMM," to address this issue. We propose a geometric concept that permits endmembers to vary freely in three dimensions, hence enhancing flexibility in all layers for any endmembers. Utilizing an innovative mathematical framework, variations can occur across three orthogonal directions (layers, pixels, or other endmembers), supporting any type of variation for all elements in the endmember signature matrix. By this concept, we have also suggested the main optimization function and two mathematical equations to improve blind hyperspectral unmixing. The results of our simulations, using both synthetic and real data sets, show that our method accurately estimates and converges better, while the mathematical analysis proves this issue.

Index Terms—Hyperspectral imaging, Blind hyperspectral unmixing, Linear Mixing Model, Extended Linear Mixing Model, Endmember variability

I. INTRODUCTION

HYPERSPECTRAL images (HSI) include various electromagnetic spectra, spanning from the visible to near-IR spectrum, as captured by spectral cameras [1],[2].

The main criterion of HSI is high spectral resolution, providing precise identification, discrimination of the scene, and subsequent analysis crucial and helpful. HSI undoubtedly has an extensive range of applications, including space exploration, earth surveillance, and military applications. HSI's capabilities to distinguish and identify various geologic layers make it extremely useful for mineral prospecting [3],[1].

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Researchers can also use it to investigate the sources of water and oil. Another possible use of HSI in diagnosing illnesses and surgery is tumor identification [4] so, the crucial role of Blind Source Separation (BSS) in the accurate detection and separation of sources is evident from its wide range of applications. As the received light collects at different wavelengths, by HSI camera, and because the scene can be far away from it, it's important to have a good signal-to-noise ratio (SNR) for accurate identification. This method requires larger pixels to get a good SNR, which reduces the detail in the images; therefore, there is a trade-off between spatial resolution and SNR in hyperspectral imaging analysis. Two ideas exist in HSU based on the aforementioned trade-off. [5]. A pixel under examination has a single pure substance, resulting in one spectral signature and wavelength. In this context, the pure pixel model is the commonly employed model. The other explanation that fits with what was just described is that a pixel is made up of different parts because the spatial resolution is not good enough and the distance between the scene and the sensor is large. Consequently, separation is necessary to ascertain the constituents of each pixel. Spectral unmixing (SU) is seen here as a problem of blind source separation (BSS) that focuses on identifying pure materials in an image, each with a unique spectral signature called an endmember (EM), along with the amounts of each material in the mixture known as fractional abundances [6]. LMM is a well-known standard model for SU, wherein the output is a linear mixture of EMs weighted by their abundance coefficients. In LMM, each EM is characterized by a distinct spectral signature. This approach is applicable when the pixel's field of vision pertains to a flat surface, where the material of interest occupies a certain region; in other words, they intermingle on a macroscopic scale [7],[8]. LMM is a well-known standard model for SU, wherein the output is a linear mixture of EMs weighted by their abundance coefficients. In LMM, each EM is characterized by a distinct spectral signature. This approach is applicable when the pixel's field of vision pertains to a flat surface, where the material of interest occupies a certain region; in other words, they intermingle on a macroscopic scale [9]. Limitations related to the use of LMM include nonlinear mixing [2], the identification of composite substances (in which EM is more accurately defined by a collection of signatures) [10] and endmember variability (whether internal or extrinsic) [11]. The incomplete modeling stated above encouraged the investigation of a substitution with improved modeling, leading to the creation of the ELMM [2].

ELMM is a physical model introduced for variability caused by illumination. It allows the variability of EM within adjacent pixels by incorporating a new scaling factor matrix within the main LMM formulation. This model has been verified by several experimental measurements; thus, EM variability can be clarified using ELMM. The formulation of ELMM, while its numerous advantages, also exposes an intrinsic multiplicative ambiguity between abundances and scaling factors [7]. Furthermore, no boundary exists that shows the precision of ELMM. Also, the model assumes all layers in one EM have the same weight, but in a real scenario, this assumption is not correct. A more deeper theoretical comprehension might be useful in the formulation [12]. Furthermore, the challenge of source separation persists in tiny nonlinear mixtures within composite materials. The present research is a response to this demand. The idea offers a novel dynamic EM variability model that allows EMs to vary freely in three dimensions, unconstrained by limitations, in order to encompass any type of EM variability. The newly developed model addresses EM variability through a three-dimensional scaling factor array, perfectly linked with each member of the EM matrix. Therefore, the ELMM formulation is being advanced to the new IELMM equation, which provides better convergence and precision in estimating the abundance matrix and EMs, along with the scaling factor matrix. We also built a mathematical framework for the aforementioned concept. Simulation results employing both real and synthetic data sets confirm the novel concept. The next few parts of this work are organized as follows: Section II defines spectral variability and its effects on HSI. The LMM and its criteria are provided in Section III. ELMM is defined in Section IV. The proposed approach and its formulation are provided in Part V. Section VI shows the simulation results using both real and synthetic data sets. Section VII includes concluding reports and future outlines.

II. SPECTRAL VARIABILITY

The spectral signatures of materials engage in changes. This behavior of materials corresponds with inherent changes in the materials, variations in lighting conditions, topographic modifications, atmospheric effects, or sensor movements [12]. Such variations are classified as intrinsic or extrinsic variations. Intrinsic variations occur due to variations in the physicochemical structure of the material, leading to variations in its spectral signature. A good illustration of this variation is the changing of chlorophyll in a cornfield through the day due to changes in sunlight, which results in a shift in its spectrum [10]. Extrinsic variations come from variations in external factors, like lighting circumstances, incidence angle, or the angle of the received signal in the sensor aperture, in connection to platform movement.

The atmosphere's effect on light signals is another observable impact. Such distinctions are known as extrinsic variations, which all lead to changes in the spectral signatures. Therefore, it is important to offer a new model that compensates for the mentioned variations in SU. The present research provides an idea that enables modifications of EMs by dynamically coupling them with a 3D scaling factor array,

which supports various types of variations more effectively. Consequently, flexibility is built into the ELMM model to enhance its capacity to adapt to variations.

III. LINEAR MIXTURE MODEL

LMM is a famous model widely used in SU. In an LMM, an observation matrix is a linear mixture of various EMs, weighted by their coefficients (abundances). Furthermore, each EM is distinguished by a unique spectral signature. This method works when the pixel in concern relates to just one material of interest; in this case, the mixing process needs to occur at a macroscopic scale. In LMM, the field of view for every pixel corresponds to a flat surface of a particular pure material [7], [8]. Notwithstanding its limitations, the method is easy to follow and adaptable. It not only defines the sources (EMs) in a clear way but also offers an appropriate first-order approximation of the mixing process [9]. LMM is typically expressed in matrix form as follows:

$$\mathbf{X} = \mathbf{S} \cdot \mathbf{A} + \mathbf{E} \quad (1)$$

where $\mathbf{X} \in \mathbf{R}^{(L \times N)}$ is the observation matrix, $\mathbf{S} \in \mathbf{R}^{(L \times P)}$ represents endmember signatures, $\mathbf{A} \in \mathbf{R}^{(P \times N)}$ denotes fractional abundances, and $\mathbf{E} \in \mathbf{R}^{(L \times N)}$ signifies Gaussian noise. $N = (m \times n)$ represents the total number of pixels in each layer. P is the quantity of EMs while L represents the number of frequency bands in hyperspectral images (layers). The model can be written in spread form for each column vector $\mathbf{x}_n \in \mathbf{R}^{L \times 1}$ for $n = 1 : N$ by [2]:

$$\mathbf{x}_n = \sum_{p=1}^P a_{pn} \cdot \mathbf{S}_p + \mathbf{e}_n \quad (2)$$

where $\mathbf{S}_p$ represents the p 'th column vector of the endmember matrix. ($a_{pn} \geq 0 (\forall p, n)$ represents abundance with no negative constraints (ANC)), indicating that none of the involved physical values are expected to be negative. ($\sum_{p=1}^P a_{pn} = 1 (\forall p, n)$ denotes the abundance sum to one constraint (ASC)), meaning that every pixel must be entirely represented by the contributions of endmembers (EMs) [2]. Despite its benefits and usefulness, LMM has several constraints. When the EM sources or their lighting conditions change, the model fails to reflect reality. Consequently, a new model is necessary. The extended linear mixing model addresses this issue.

IV. EXTENDED LINEAR MIXING MODEL

In the extended linear mixing model, EMs should be allowed to vary throughout adjacent pixels, while the mixing process remains linear. This variability is defined by the scaling (weighting) of particular reference endmembers (ref EMs) organized in a ref EM matrix denoted as $\mathbf{S}_0 \in \mathbf{R}^{(L \times P)}$ [2]. Therefore, the variation of any EM can be reflected by weighting the reference EM matrix with a diagonal scaling factor matrix $\mathbf{\Psi} \in \mathbf{R}^{(P \times P)}$. Therefore, the p 'th scaling factor of the matrix affects any p 'th column of the reference EM matrix. The pixel-wise variation of each EM can be stated as follows [2]:

$$\mathbf{x}_k = \sum_{p=1}^P a_{pk} f_{pk}(\mathbf{S}_p) + \mathbf{e}_k \quad (3)$$

Here, a_{pk} indicates the abundance coefficient for pixel k and material p , while f_{pk} reveals the variability of the EMs. We assume that $f_{pk}(\mathbf{S}_p) = \psi_{pk}\mathbf{S}_{0p}$, implying that the weighting of expected reference EMs $\mathbf{S}_0$ is responsible for the spectral variability. In $\mathbf{S}_0$, each column corresponds to an EM vector $\mathbf{S}_{0p}$, for $p = 1 : P$. Equation (3) can be reformulated as:

$$\mathbf{x}_k = \sum_{p=1}^P a_{pk}\psi_{pk}\mathbf{S}_{0p} + \mathbf{e}_k = \mathbf{S}_0\mathbf{\Psi}_k\mathbf{a}_k + \mathbf{e}_k \quad (4)$$

where $\mathbf{\Psi}_k \in \mathbf{R}^{(P \times P)}$ gathers all scaling factors of all EMs, which is a diagonal matrix with values $\psi_{pp} > 0$ and $\mathbf{S}_0 \in \mathbf{R}^{(L \times P)}$ [7]. As it can be seen in (4) the EMs in this model can vary between neighboring pixels, meaning EMs can vary by scaling factor to get a deviation in accordance with the p 'th number of EMs. Also, the ASC could be confirmed because coefficients are moved to $\mathbf{\Psi}$ matrix. By this model, some constraints referred to in LMM are solved. The model has been validated by many experimental measurements [2]. Although the EM variability problem is nearly modeled by ELMM, during usage, some limitations occur as mentioned in sec I. Our proposed model is presented as a response to this demand .

V. PROPOSED METHOD

This section suggests a novel dynamic model that optimally matches the physics of endmember variation. The model allows EMs to vary freely; it also takes into account any variations in the proposed formulation. An innovative 3D scaling factor array, which we offered, is connected element-wise to the offered reference endmembers which we converted to 3-D array as seen in Fig.1. In this paradigm, each element in the new 3-D source array can vary independently throughout pixels, layers, or EMs. Following our suggestion, the diagonal scaling factor matrix $\mathbf{\Psi} \in \mathbf{R}^{(P \times P)}$ in ELMM is changed into a 3-D $(L \times P \times N)$ $\mathbf{\Psi}$ array with all positive numbers, as shown in Fig.1. The proposed approach specifies the 3D source array "S" for estimation in the following manner:

$$\mathbf{S} = \mathbf{S}_{03D(L \times P \times N)} \odot \mathbf{\Psi}_{(L \times P \times N)} \quad (5)$$

Where $\mathbf{S}_{03D}$ is N times the repetition of $\mathbf{S}_{0(L \times P)}$ with regard to N layers (as seen in Fig. 1), and the notation $\odot$ is defined as the Schur-Hadamard or element-wise product. We connect each element of the proposed array $\mathbf{\Psi}$, element by element, to the new shaped array of "S" (as defined above) by employing the Hadamard Product. Section (A) describes the eventual development of the ELMM formulation into the new IELMM formulation, and in section (B) the main optimization function is discussed, based on this concept.

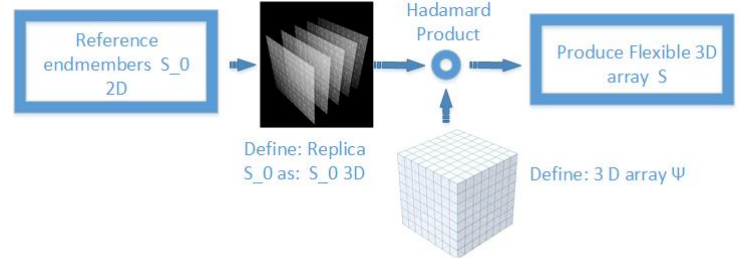


Fig. 1: Schematic diagram of the proposed geometric idea

A. Problem Formulation

Based on above discussions, we propose the Enhanced ELMM (IELMM) as follows:

$$\begin{aligned} \mathbf{x}_k &= (\mathbf{S}_{0(L \times P)} \odot \mathbf{\Psi}_{(L \times P)})_k \mathbf{a}_k + \mathbf{e}_k \quad (6) \\ &= \mathbf{S}_k \mathbf{a}_k + \mathbf{e}_k \\ k &= 1, \dots, N \\ \mathbf{X} &= [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N] \\ \mathbf{A} &= [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_N] \\ \mathbf{E} &= [\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_N] \end{aligned}$$

where $\mathbf{X} \in \mathbf{R}^{(L \times N)}$ is input observation matrix and $\mathbf{x}_k$ is k 'th column vector of $\mathbf{X}$ for $k = 1 : N$. $\mathbf{\Psi}_k$ denotes the k 'th surface of the array, and $\mathbf{S}_{0k}$ denotes the k 'th surface of the 3-D S array which is involved. $\mathbf{A} \in \mathbf{R}^{(P \times N)}$ is an abundance matrix, and $\mathbf{a}_k$ is the k 'th column vector of $\mathbf{A}$. $\mathbf{E} \in \mathbf{R}^{(L \times N)}$ is additive white Gaussian noise, and $\mathbf{e}_k$ is k 'th column vector of $\mathbf{E}$. $N = (m \times n)$ represents the total number of pixels in the image, where (m, n) denotes the dimensions of the image in the spatial domain in each layer. The starting point in the blind hyperspectral unmixing method involves deriving the ref EM matrix ($\mathbf{S}_0$). It can be acquired by any of the reference EM extraction strategies outlined in [13]. The Minimum Volume Simplex Analysis (MVSA) method, N-FINDR, and vertex component analysis (VCA) are generally useful techniques employed in the extraction of ref EMs in this field. In our research, which builds upon Nascimento's study in [13] and its comparative analysis of the aforementioned algorithms, we employed the VCA algorithm to extract reference endmembers as an initial phase of the proposed blind HSU technique.

B. Main Optimization Function

In this section, the main optimization function of the proposed blind HSU method is provided. Our main optimization

function in terms of three parameters, $\mathbf{S}$, Ψ , and $\mathbf{A}$ is:

$$F(\mathbf{S}, \Psi, \mathbf{A}) = \frac{1}{2} \sum_{k=1}^N (\|\mathbf{x}_k - \mathbf{S}_k \mathbf{a}_k\|_2^2 + \lambda_S \|\mathbf{S}_k - (\mathbf{S}_0_{(L \times P)} \odot \Psi_{(L \times P)})_k\|_F^2 + \text{reg}(\Psi_k)) + \text{reg}(\mathbf{A}) \quad (7)$$

$$\begin{aligned} \text{reg}(\mathbf{A}) &= \mathbf{I}_R^+(\mathbf{A}) + \mu^T (\mathbf{A}^T \mathbf{1}_P - \mathbf{1}_N) + \lambda_A SR(\mathbf{A}) \\ &= \lambda_A (\|\mathbf{H}_v(\mathbf{A})\|_{2,1} + \|\mathbf{H}_h(\mathbf{A})\|_{2,1}) + \mathbf{I}_R^+(\mathbf{A}) \end{aligned} \quad (8)$$

$$\begin{aligned} &+ \mu^T (\mathbf{A}^T \mathbf{1}_P - \mathbf{1}_N) \\ \text{reg}(\Psi_k) &= \frac{\lambda_\psi}{2} (\|\mathbf{H}_{\psi v}(\Psi_k)\|_F^2 + \|\mathbf{H}_{\psi h}(\Psi_k)\|_F^2) \end{aligned} \quad (9)$$

Where $\mathbf{S}_k$ indicates the k' th surface of the 3-D source array $\mathbf{S}$, which is expressed as $(L \times P)$. $\mathbf{H}_h$ and $\mathbf{H}_v$ are $(P \times N)$ horizontal and vertical gradient operators in adjacent pixels acting individually on each material. $\mathbf{H}_{\psi h}$ and $\mathbf{H}_{\psi v}$ are $(L \times P)$ horizontal and vertical gradient operators, respectively, which operate on each surface (layer) of the array with respect to the index k (number of pixels). $\mu^T \in \mathbf{R}^N$ is a column vector of Lagrange multipliers for ASC constraint. λ_ψ and λ_A are regularization parameters that affect the convergence rate in iterative loops [2]. $\mathbf{I}_R^+$ is a $(P \times N)$ indicator function of the non-negative orthant. $\mathbf{1}_{(\cdot)}$ implies a column vector including ones, with its length reflected in its index. $SR(\mathbf{A})$ demonstrates the spatial regularization of abundance ($\mathbf{A}$). According to references [14] and [15], we can replace the mixed norm $L(2,1)$ with $L(1,1)$ because of the total variation penalization (TV) mentioned in [15]. Therefore, as shown in equations (8), we carry out the spatial regularization for each abundance map using the $L(1,1)$ norm. We applied first-order neighborhoods for gradient computations [2].

The solution of the proposed method is achieved by minimizing the main optimization function in (7) with respect to each parameter $\mathbf{S}$, Ψ , $\mathbf{A}$ individually using a block coordinate descent (BCD) procedure. The problem is not convex concerning all parameters simultaneously, but it is convex with respect to each individual parameter. We solve each sub-problem and seek the optimal point using iterative processes within the comprehensive BCD loop, as illustrated in the pseudo code of Algorithm 1.

1) analyze the optimization w.r.t $\mathbf{S}$: To estimate EM signatures ($\hat{\mathbf{S}}$), we must minimize equation (7) w.r.t $\mathbf{S}$ as follows:

$$\hat{\mathbf{S}} = \underset{\mathbf{S}}{\text{argmin}} F(\mathbf{S}, \Psi, \mathbf{A}) \text{ w.r.t : } \mathbf{S}$$

$$\begin{aligned} \hat{\mathbf{S}}_k &= \underset{(\mathbf{S}_k)}{\text{argmin}} \left(\frac{1}{2} \sum_{k=1}^N (\|\mathbf{x}_k - \mathbf{S}_k \mathbf{a}_k\|_2^2 + \lambda_S \|\mathbf{S}_k - (\mathbf{S}_0_{(L \times P)} \odot \Psi_{(L \times P)})_k\|_F^2) \right); k = 1, \dots, N \end{aligned} \quad (10)$$

The optimization problem in (10) w.r.t $\mathbf{S}_k$ has a closed form solution; so, by using derivative w.r.t. $(\mathbf{S}_k)$, the solution can be expressed as:

$$\hat{\mathbf{S}}_k \leftarrow (\mathbf{x}_k \mathbf{a}_k^T + \lambda_S (\mathbf{S}_0 \odot \Psi_k)) (\mathbf{a}_k \mathbf{a}_k^T + \lambda_S \mathbf{I}_P)^{-1} \quad (11)$$

Since EM signatures must be positive, the estimated $(\mathbf{S}_k)$ should be confined to non negative orthant $\mathbf{R}^{+(L \times P)}$ in each k by thresholding the results with zero to accept only positive values.

Algorithm1: Iterative BCD pseudo code to find local minimum of the main optimization function $F(\mathbf{S}, \Psi, \mathbf{A})$

Inputs: $\mathbf{X}$ is observation matrix and $\mathbf{S}_0$ is ref EM matrix.
Outputs: $\hat{\mathbf{S}}$, $\hat{\Psi}$ and $\hat{\mathbf{A}}$.
Initialization: initialize $\mathbf{S}$ as: $S_{i,j,k} > 0$ and $\mathbf{A}$ as: $a_{i,j} > 0$ with (ASC) constraint and initialize Ψ .

↓ **while** Iterative BCD criteria is not satisfied, **do:**

$$\downarrow \hat{\mathbf{S}} = \underset{\mathbf{S}}{\text{argmin}} F(\mathbf{S}, \Psi, \mathbf{A}) \text{ w.r.t : } \mathbf{S}$$

$$\downarrow \hat{\Psi} = \underset{\Psi}{\text{argmin}} F(\mathbf{S}, \Psi, \mathbf{A}) \text{ w.r.t : } \Psi$$

$$\downarrow \hat{\mathbf{A}} = \underset{\mathbf{A}}{\text{argmin}} F(\mathbf{S}, \Psi, \mathbf{A}) \text{ w.r.t : } \mathbf{A}$$

end

2) Analyze the optimization w.r.t Ψ : To estimate the scaling factor coefficients we must minimize equation (7) w.r.t Ψ as follows:

$$\hat{\Psi} = \underset{\Psi}{\text{argmin}} F(\mathbf{S}, \Psi, \mathbf{A}) \text{ w.r.t : } \Psi$$

$$\begin{aligned} \hat{\Psi}_k &= \underset{(\Psi_k)}{\text{argmin}} \left(\frac{1}{2} \sum_{k=1}^N (\lambda_S \|\mathbf{S}_k - (\mathbf{S}_0_{(L \times P)} \odot \Psi_{(L \times P)})_k\|_F^2 + \frac{\lambda_\psi}{2} (\|\mathbf{H}_{\psi h}(\Psi_k)\|_F^2 + \|\mathbf{H}_{\psi v}(\Psi_k)\|_F^2)) \right); k = 1, \dots, N \end{aligned} \quad (12)$$

The optimization with respect to Ψ_k is as follows:

$$\begin{aligned} \frac{dF(\mathbf{S}, \Psi, \mathbf{A})}{d\Psi_k} &= -\lambda_S \mathbf{S}_0 \odot (\mathbf{S}_k - (\mathbf{S}_0_{(L \times P)} \odot \Psi_{(L \times P)})_k) + \\ \lambda_\Psi (\mathbf{H}_{\psi h}^T \mathbf{H}_{\psi h}(\Psi_k) + \mathbf{H}_{\psi v}^T \mathbf{H}_{\psi v}(\Psi_k)) &= 0 \end{aligned} \quad (13)$$

$$\begin{aligned} \lambda_S \mathbf{S}_0 \odot (\mathbf{S}_0_{(L \times P)} \odot \Psi_{(L \times P)})_k &= -\lambda_\Psi (\mathbf{H}_{\psi h}^T \mathbf{H}_{\psi h}(\Psi_k) + \mathbf{H}_{\psi v}^T \mathbf{H}_{\psi v}(\Psi_k)) + \lambda_S \mathbf{S}_0 \odot \mathbf{S}_k \equiv \mathbf{U}^n \\ \rightarrow (S_0 \odot \Psi_k)_{i,j}^{n+1} &= \frac{U_{i,j}^n}{\lambda_S S_{0i,j}} \\ \rightarrow (\hat{\Psi}_k)_{i,j}^{n+1} &= \frac{U_{i,j}^n}{\lambda_S (S_0)_{i,j}^2}; \text{ for } (k = 1, 2, \dots, N) \end{aligned} \quad (14)$$

Equation (14) is an iterative equation, and n is the number of iterations. Using the Ψ array set in the initialization step of algorithm 1, the optimum Ψ can be obtained by the iterative equation in (14).

hint: For researchers following this article, we remark that in problems involving the Hadamard product in conjunction with matrix multiplication, the derivatives differ significantly from standard matrix calculations. We articulate the derivative rule in this context as follows:

$$\begin{aligned}
D &= \frac{\partial \|\mathbf{A} \odot \mathbf{X} - \mathbf{B}\|_{\mathbf{F}}^2}{\partial \mathbf{X}} = ? \\
\text{def : } \mathbf{A} &= [a_{ij}]_{n \times m}; \mathbf{X} = [x_{ij}]_{n \times m}; \mathbf{B} = [b_{ij}]_{n \times m} \\
\mathbf{C} &= [a_{ij}x_{ij} - b_{ij}]_{n \times m}; \rightarrow \\
\frac{\partial (\sum_{i'j'} (a_{i'j'}x_{i'j'} - b_{i'j'})^2)}{\partial x_{i'j'}} &= \sum_{i'j'} 2a_{i'j'}(a_{i'j'}x_{i'j'} - b_{i'j'}) \\
\rightarrow \mathbf{D} &= 2\mathbf{A} \odot (\mathbf{A} \odot \mathbf{X} - \mathbf{B}) \\
\rightarrow \frac{\partial (\|\mathbf{S}_k - (\mathbf{S}_0 \odot (\mathbf{L} \times \mathbf{P}) \odot \Psi)_{(L \times P)k}\|_F^2)}{\partial \Psi_k} &= \\
-2\mathbf{S}_0 \odot (\mathbf{S}_k - (\mathbf{S}_0 \odot \Psi)_k) & \quad (16)
\end{aligned}$$

3) Analysis the optimization w.r.t $\mathbf{A}$:

To estimate $\mathbf{A}$ it's needed to minimize equation (7) w.r.t $\mathbf{A}$ as follows:

$$\begin{aligned}
\hat{\mathbf{A}} &= \underset{\mathbf{A}}{\operatorname{argmin}} F(\mathbf{S}, \Psi, \mathbf{A}) \text{ w.r.t : } \mathbf{A} \\
\hat{\mathbf{A}} &= \underset{\mathbf{A}}{\operatorname{argmin}} \frac{1}{2} \sum_{k=1}^N (\|\mathbf{x}_k - \mathbf{S}_k \mathbf{a}_k\|_2^2) + \mathbf{I}_{R^+}(\mathbf{A}) \\
&+ \mu^T (\mathbf{A}^T \mathbf{1}_P - \mathbf{1}_N) + \lambda_A (\|\mathbf{H}_h(\mathbf{A})\|_{2,1} + \|\mathbf{H}_v(\mathbf{A})\|_{2,1}) \quad (17)
\end{aligned}$$

To address this sub-problem, an approach utilizing Alternating Direction Method of Multipliers (ADMM) is proposed. The method is employed in a manner analogous to that described in [16] and [2]. In ADMM, the intricate issue is decomposed to simpler sub-problems that possess closed-form solutions. Each sub-problem is addressed individually, sequentially. The sequence of any outcome is subsequently applied to others iteratively to achieve the main desired result. This technique can decouple spatial domain optimization from frequency domain optimization, akin to the method described in [15]. In our study, ten unknown parameters (five auxiliary Lagrange parameters (Ω_i , $i = 1, 2, \dots, 5$) and five Lagrangian multipliers ($\mathbf{m}_i$, $i = 1, 2, \dots, 5$) are defined, linked together, computed by various methods, and iteratively updated in sequence to derive the optimal $\mathbf{A}$.

We like to formulate equation (17) and decompose it into a simpler subproblem for enhanced comprehension and implementation. *No nobility could be observed in the method of the solution*, but it aims to be accessible and easy to apply.

4) Problem Formulation: To solve equation (17), we must define it using new parameters while maintaining the (ASC) and (ANC) constraints. We reframe the issue as the FCLSU problem and apply the FCLSU rule, as stated in [17], to resolve it effectively. It is necessary to alternate between the vectorized form of the equation and the matrix form depending on the circumstances. The matrix form is a concise and seemingly convenient representation, while both results are equivalent. During the optimization process, we apply whichever method is most suitable for the circumstances at hand. We designate the splitting variables in equation (17) as new variables ($\Omega_1, \dots, \Omega_5$) to serve as auxiliary parameters.

Constructing the matrix representation of equation (17) is:

$$\begin{aligned}
\hat{\mathbf{A}} &= \underset{\mathbf{A}}{\operatorname{argmin}} \frac{1}{2} \sum_{k=1}^N \|\mathbf{X}_k - \omega_{1k}\|_2^2 + \lambda_A (\|\Omega_3\|_{2,1} + \|\Omega_4\|_{2,1}) \\
&+ \mathbf{I}_{R^+}^{+}(\Omega_5) + \mu^T \mathbf{A}^T (\mathbf{1}_P - \mathbf{1}_N); \text{ s.t : } \\
\Omega_1 &= [\mathbf{S}_1 \mathbf{a}_1 \dots \mathbf{S}_N \mathbf{a}_N]; \Omega_2 = \mathbf{A}; \Omega_3 = \mathbf{H}_h(\Omega_2); \\
\Omega_4 &= \mathbf{H}_v(\Omega_2); \Omega_5 = \mathbf{A} \quad (18)
\end{aligned}$$

Where " $\mathbf{X}_k$ " is a vector indicate k' th column of observation matrix " $\mathbf{X}$ " and $(\omega_1, \omega_2 \dots \omega_5)$ are vectorized version of $(\Omega_1 \Omega_2 \dots \Omega_5)$; So ω_{ik} is k' th column vector of Ω_i matrix"

Following problem (17) in the frame work of ADMM, yields:

$$\begin{aligned}
(\hat{\Delta}, \hat{\Theta}) &= \underset{\Theta}{\operatorname{argmin}} f(\Theta) \text{ s.t } \Upsilon \Delta + \Phi \Theta = 0 \quad (19) \\
\text{where : } \Delta &= \mathbf{a} = \operatorname{vec}(\mathbf{A}), \Theta = [\omega_1 \omega_2 \dots \omega_5]^T \\
f(\Theta) &= \frac{1}{2} \|\mathbf{x} - \omega_1\|_2^2 + \lambda_A (\|\operatorname{vec}^{-1}(\omega_3)\|_{2,1} + \\
&\|\operatorname{vec}^{-1}(\omega_4)\|_{2,1} + \mathbf{I}_{R^+}(\omega_5) + \mu^T (\mathbf{K} \mathbf{a} - \mathbf{1}_N)) \\
\Upsilon &= [\Sigma^T \quad \mathbf{I}_{P \times N} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{I}_{P \times N}] \\
\Phi &= \begin{bmatrix} -\mathbf{I}_{L \times N} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{I}_{P \times N} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_h & -\mathbf{I}_{P \times N} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_v & \mathbf{0} & -\mathbf{I}_{P \times N} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{I}_{P \times N} \end{bmatrix}
\end{aligned}$$

In this formulation $\mathbf{H}_h, \mathbf{H}_v \in \mathbf{R}^{P \times N} \rightarrow \mathbf{R}^{P \times N}$ are horizontal and vertical gradient operators. $\mathbf{I}_{(\cdot)}$ denotes the identity matrix which its length reflected in its index. $\Sigma \in \mathbf{R}^{LN \times PN}$. $\mathbf{K}$ is $N \times PN$ matrix of the linear operators summing the entries of a corresponding to $\mathbf{a}$ the same pixel and putting each of these sums in one entry of a vector.

The ADMM technique involves rewriting the constrained problem from Equation (19) as an unconstrained problem (equation (20)) using an Augmented Lagrangian (AL), and then minimizing it step by step for each variable, including the Lagrange Multipliers in the AL. In the revised equation, $\mathbf{m}$ represents the Lagrange multiplier in vector syntax, or $\mathbf{M}$ in matrix form. ρ is the barrier parameter that weights the AL equation. The AL equation could eventually be minimized depending on $\Delta, \Theta, \mu, \mathbf{m}$ as:

$$\begin{aligned}
\mathcal{L}(\Delta, \mu, \Theta, \mathbf{m}) &= f(\Theta) + g(\Delta) + \frac{\rho}{2} (\|\Upsilon \Delta + \Phi \Theta - \mathbf{m}\|_2^2 - \\
\|\mathbf{m}\|_2^2) &= \frac{1}{2} \|\mathbf{x} - \omega_1\|_2^2 + \lambda_A (\|\operatorname{vec}^{-1}(\omega_2)\|_{2,1} + \|\operatorname{vec}^{-1}(\omega_3)\|_{2,1} \\
&+ \mathbf{I}_R^+(\omega_4) + \mu^T (\mathbf{K} \Delta - \mathbf{1}_N) + \frac{\rho}{2} \|\Sigma \Delta - \omega_1 - \mathbf{m}_1\|_2^2 + \\
\frac{\rho}{2} \|\Delta - \omega_2 - \mathbf{m}_2\|_2^2 &+ \frac{\rho}{2} \|\mathbf{H}_h \omega_2 - \omega_3 - \mathbf{m}_3\|_2^2 + \\
\frac{\rho}{2} \|\mathbf{H}_v \omega_2 - \omega_4 - \mathbf{m}_4\|_2^2 &+ \frac{\rho}{2} \|\Delta - \omega_5 - \mathbf{m}_5\|_2^2 - \frac{\rho}{2} \|\mathbf{m}\|_2^2 \quad (20)
\end{aligned}$$

Now equation (20) is an unconstrained problem, for which it could be possible to use the FCLSU algorithm to estimate parameters separately. The unknowns $\Omega_1, \dots, \Omega_5$ are auxiliary wanted parameters, and $\mathbf{m}_1, \dots, \mathbf{m}_5$ are wanted multipliers. Finally, the parameter $\mathbf{A}$ or $\Delta = \mathbf{a} = \operatorname{vect}(\mathbf{A})$ can be extracted during the process by following the below

TABLE I: 5 standard reference endmembers used for experiments with synthetic data set 2

RefEMs	S ₀₁	S ₀₂	S ₀₃	S ₀₄	S ₀₅
L ₁	0.0130	0.0300	0.2650	0.0950	0.2170
L ₂	0.0190	0.0310	0.2710	0.0960	0.2220
L ₃	0.0250	0.0370	0.2680	0.0990	0.2250
⋮	⋮	⋮	⋮	⋮	⋮
L ₁₆₁	0.0570	0.0850	0.1520	0.1800	0.4560
L ₁₆₂	0.0400	0.0610	0.1360	0.1680	0.4780

chain:

initialization: $\Delta, \mu, \Theta, \mathbf{m}, 0 \leq \rho \leq 1$

Inputs: $\mathbf{X}, \mathbf{S}$

output: $\hat{\mathbf{A}}$

A) $\Delta, \mu \leftarrow \argmin \mathcal{L}(\Delta, \mu, \Theta, \mathbf{m})$ w.r.t Δ

B) $\Theta \leftarrow \argmin \mathcal{L}(\Delta, \mu, \Theta, \mathbf{m})$ w.r.t Θ

C) $\mathbf{m} \leftarrow \mathbf{m} - (\Upsilon \Delta + \Phi \Theta)$

D) If convergence occurred stop else, go to (A).

hint: Researchers referencing this article will find that we addressed this sophisticated problem utilizing the methodology outlined in [18], which serves as extra information to reference [2].

VI. EXPERIMENTS ON REAL AND SYNTHETIC DATA SETS

We execute two experiments using real and synthetic datasets to demonstrate the effectiveness of the proposed method. The outcomes of the suggested IELMM-based methodology are presented and compared with the ELMM-based method outlined in [2]. In the first test, a synthetic data set is generated and utilized for HSU employing both ELMM and IELMM methodologies. The specifics of the synthetic data set experiment and comparative analysis are outlined in section VI-A. In the real data set experiment, we utilize a real HSI data set, depicting the Pavia University Campus in Italy. This data set is general data that is used in several experiments. We conducted two experiments with the real data set to compare the speed during simulation and the manner of obtaining the final results in converging the response in each methodology. The proposed approach for HSU, grounded on the IELMM model, is simulated using this actual dataset, and the outcomes are compared with the prior method. The results of two experiments and its analysis are detailed in section VI-B.

A. Experiments on Synthetic data sets

Two primary data sets are required to initiate the experiment. We need the observation matrix ($\mathbf{X}$), and reference EM signatures ($\mathbf{S}_0$). In ref EMs ($\mathbf{S}_0$), we utilize five standard spectral signatures from the remote sensing laboratory at Tehran University, Tehran, Iran, as presented in Table I (this data is accessible at the address (<https://rslab.ut.ac.ir/data>)). The 3-D scaling factor array Ψ is generated using a Gaussian distribution that ranges from 0.4 to 1.35, based on what we learned from realistic simulations, as well as reference [2]. Each surface of this array at the k'th pixel is multiplied element-wise with the reference EM matrix to generate the EM

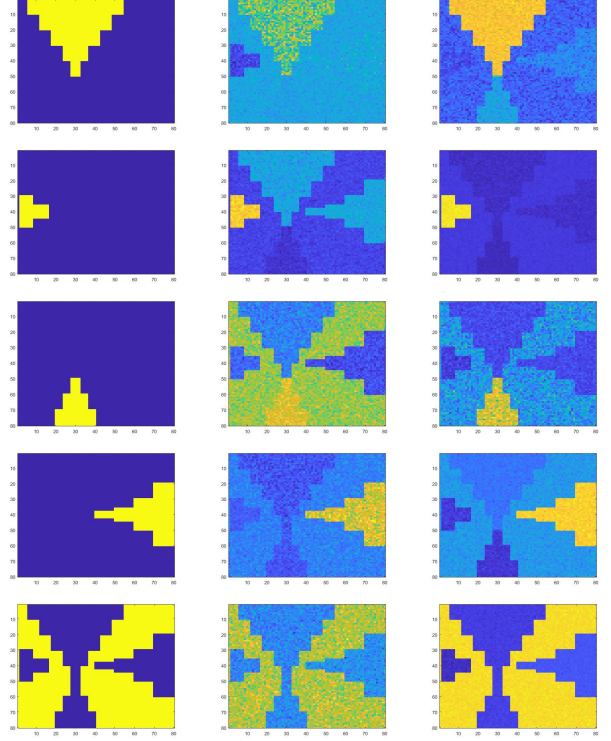


Fig. 2: Results of the experiment utilizing synthetic data set. The left column displays pure true abundance maps for five endmembers, the middle column presents estimated maps based on the ELMM model, and the right column shows estimated maps derived from the proposed IELMM technique.

signature $\mathbf{S}_k$ (for $k = 1 : N$) as refereed in equation (5) and (6) and illustrated in Fig. 1 1. We have generated five synthetic abundance maps, corresponding to five EMs, according to ANC and ASC constraints, to create the abundance matrix ($\mathbf{A}$), as shown in the first column of Fig.2. The observation matrix ($\mathbf{X}$) is subsequently generated based on equation ((6)). The dimensions of the observation matrix are designated as $162 \times (80 \times 80)$, denoted as $L \times (m \times n)$ in the spatial domain. We employed two methodologies, ELMM and our proposed IELMM method, on these datasets to estimate $\hat{\mathbf{A}}$, $\hat{\mathbf{S}}$, and $\hat{\Psi}$ and compared the results of each method with their original values to evaluate the efficiency of both approaches. To establish a quantitative comparison, three criteria are defined as follows:

$$A_{RMSE} = \frac{1}{N} \sum_{k=1}^N \sqrt{\frac{1}{P} \sum_{p=1}^P (a_{pk \text{ true}} - \hat{a}_{pk})^2} \quad (21)$$

$$\Psi_{RMSE} = \frac{1}{N} \sum_{k=1}^N \sqrt{\frac{1}{LP} \|(\Psi_{k \text{ true}} - \hat{\Psi}_k)\|_F^2} \quad (22)$$

$$S_{RMSE} = \frac{1}{N} \sum_{k=1}^N \sqrt{\frac{1}{LP} \|\mathbf{S}_{k \text{ true}} - \hat{\mathbf{S}}_k\|_F^2} \quad (23)$$

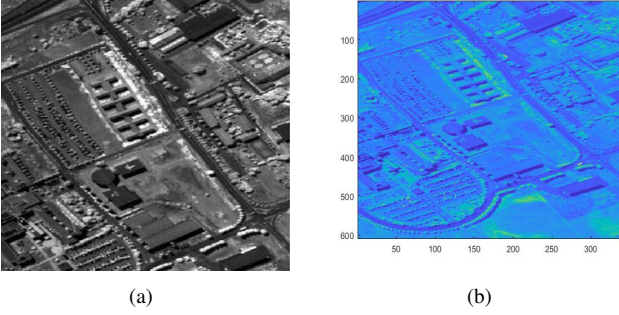


Fig. 3: Two images of Pavia university campus (Full image), (a) High spatial resolution black and white image; (b) One sample of HSI(86'th layer)

Results of the quantitative analysis are illustrated in table II. The abundance matrix is estimated with more precision using the suggested method, as demonstrated in table II. Regarding the estimation of (Ψ) , there exists no other method to evaluate the 3D (Ψ) array for comparison. The proposed method demonstrates superior precision in extracting distinct sources as well. Also, the abundance maps estimated using the ELMM method and the new IELMM approach are displayed in the middle and right columns of Fig.2, respectively. ELMM demonstrates lower efficiency relative to IELMM, particularly in edge detection and overall image sharpness, as evidenced by the estimated abundance maps of EM1, EM3, and EM4 under identical conditions (up: EM1 to down: EM5). The proposed model takes into account changes in EM in every direction through layers, pixels, or internal variations while the math behind it is set up to handle this, ensuring precise parameter estimation.

B. Experiments on real data sets

Two experiments are done with real hyperspectral data, showing Pavia University Campus in Italy, which is presented in Fig.3(a) as a black and white image and in 3(b) as the 86th layer of the hyperspectral image. We conducted the first experiment to demonstrate how quickly we can estimate the abundance matrix $\mathbf{A}$, showing the progress in each step of the BCD stages of algorithm 1 until we reach the final result for both methods. In the second experiment, we select a cube of data from a different portion of this dataset as the observation matrix and compare the results.

TABLE II: Results of quantitative analysis for experiment with synthetic data set

Algorithm	IELMM	ELMM
A_{rmse}	0.0248	0.0643
Ψ_{rmse}	0.0953	—
S_{rmse}	0.0173	0.0242

1) *First experiment with Real data set:* The Pavia HSI data set has a total dimension of $103 \times (610 \times 340)$ as $L \times (m \times n)$ where, $N = (n \times m)$. For the first experiment, we choose a cube of data with $24 \times (90 \times 110)$ dimensions in range (80 : 103, 20 : 110, 120 : 230) from original data (northeast

part of main image) as observation matrix $\mathbf{X}$. The other needed data is EM signature matrix $\mathbf{S}_0$. It can be acquired by any of the reference EM extraction strategies outlined in [13]. The MVSA method, N-FINDR, and VCA are generally useful techniques employed in the extraction of ref EMs in this field. As discussed in section V-A, in our research, which builds upon Nascimento's study in [13] and its comparative analysis of the aforementioned algorithms, we employed the VCA algorithm to extract $\mathbf{S}_0$ as an initial phase of the proposed blind HSU technique. We choose 4 endmembers ($P = 4$) arbitrarily in this experiment. The popular procedure to find the number of EMs in intrinsic dimensional estimation is the Hysime Algorithm, as referred to in [19] (it wasn't in our research scopes). We show the estimated outputs of both algorithms based on ELMM and IELMM for the same endmember in each iteration to compare the convergence capability of each method. Estimated abundance maps of EM2 in the first 3 iterations are shown in Fig4. The left column of Fig. 4 (from top to bottom) has estimated abundance maps based on the ELMM model in each iteration loop, and the right column has estimated ones based on the proposed approach. It is evident that the proposed approach exhibits superior convergence capability. It is obvious that the proposed approach has acceptable output even in the second iteration loop, which is not seen in ELMM results. Furthermore, estimated abundance maps of three different endmembers are shown in Fig. 5. The first row of Fig. 5 shows the results based on the ELMM model, and the second row of Fig. 5 demonstrates the results based on the proposed IELMM approach. The left column of Fig. 5 shows the first EM (green land and trees), the middle column shows the second EM (asphalt area), and the right column represents the third EM (metal sheet roofs). The proposed method clearly yields better results, particularly in detecting edges and borders.

During the simulation, we have observed experimentally the importance of regularization parameters in obtaining good results, which is referred to in the supplementary material in [18]. By tuning the correct range on $\lambda_\Psi, \lambda_A, \lambda_S$, faster convergence and more precise estimation could be obtained.

2) *Second experiment with real data set :* In the second experiment, we select a data cube with $94 \times (61 \times 131)$ dimensions, labeled $L \times (m \times n)$, from the original HSI data, focusing on the range (10 : 103, 200 : 260, 90 : 220) to create the observation matrix $\mathbf{X}$, which is shown in the upper middle part of the main image. We employ the VCA approach once more as an EM extraction algorithm to compute $\mathbf{S}_0$. Three endmembers are selected ($P = 3$) for this experiment. Fig.6(a) and Fig.6(b) display the RGB image of the selected area and its approximate ground truth. The spectrum reflectance graph for five EMs is illustrated in Fig.6(c). Fig.6(d) and Fig.6(e) show the black and white image near the region of interest (ROI) along with the HSI input image from the 86th layer of this portion. Estimated abundance maps of 3 different endmembers, including shadow, meadows, and metal sheets, which are identified by yellow lighted regions, are shown in Fig.6(i), 6(j), and 6(k), based on their distribution in every area for the proposed method. Furthermore, 6(f), 6(g) and 6(h) respectively

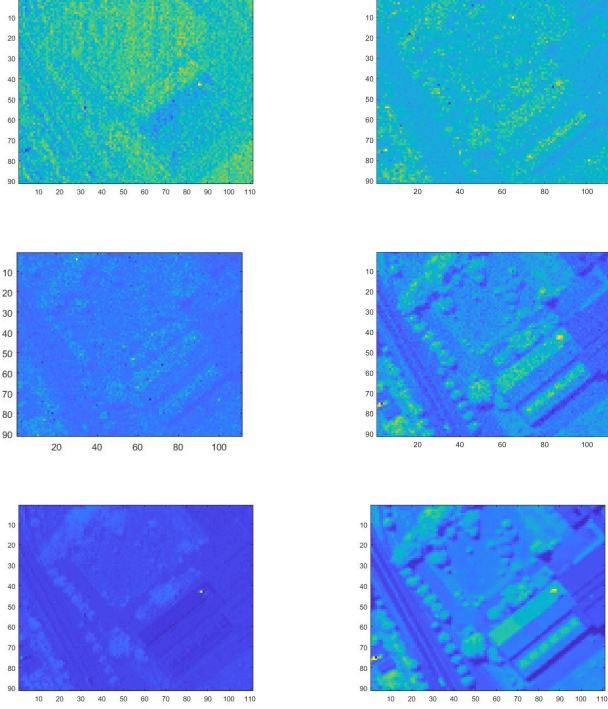


Fig. 4: Estimated abundance maps of EM2 in the first 3 iterations, for the first experiment with real data set. Left column are estimated ones based on ELMM model, and right column are estimated ones based on the proposed IELMM approach

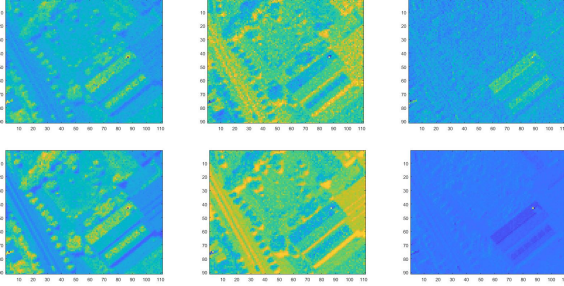


Fig. 5: Estimated abundance maps of 3 endmembers for the first experiment with real data set. First row: results of ELMM model, and second row: results of the proposed IELMM approach. Left column: green land, middle column: asphalt area, and right column: metal sheet roofs.

display the estimated abundance maps based on the ELMM approach. By focusing on the results, it's clear that separate sources adhere to their ground truth. This demonstrates the accuracy of estimation in both methods. While emphasizing the results, it appears that the ELMM in shadow regions exhibits superior resolution responses. nonetheless, it lacks any discrepancies to rectify the answer. Certain yellow areas in Fig. 6(f), which show bare soil, are incorrectly labeled as shadow in EM1 when compared to the real image in Fig. 6(a). Additionally, shaded areas do not exist in the right

corners (the findings for the main part and left corners are accurate). In the middle maps, the IELMM response in Fig. 6(j) is better at accurately showing meadow areas and matches the ground truth more closely than Fig. 6(g). In the third map (metal sheet areas), although both estimated maps demonstrate adequate responses in estimating, the IELMM response in Fig. 6(k) exhibits superior alignment with the actual metal sheet regions depicted in Fig. 6(a), and Fig. 6(b). Based on the above results, it can be concluded that the estimations derived from the suggested method exhibit superior precision in finding the outcomes.

VII. CONCLUSION

This research presents an innovative approach to address the HSU problem. Estimating HSU outcomes using this model represents a more sophisticated problem, but we have completely solved it. We ground our study in the ELMMM method to preserve the framework of linearity, but we consider a novel geometric concept that enhances HSU in the sense of EM variability. The variability of EM in the HSU problem is solved by introducing a new array, which we link to the EM signature matrix as a source of variation. We propose changing the reference EM signature matrix into a 3-D array and then connecting it to the new array using the Hadamard product to make the EM signatures more flexible, allowing for free variation in three different directions. Additionally, we present a new main optimization function that better reflects the real situations of EM variation, resulting in two new equations that provide more accurate estimates of HSU problem outcomes. Additionally, quantitative variation of any pixel in each layer including changes in angle of observation, manner of mixing and scene variation could be identified and their changes are derived numerically. Experiments utilizing both actual and synthetic datasets demonstrate the superiority of the suggested approach through quantitative analysis of the results and qualitative distinctions in outcomes among the different input matrices. Throughout the simulations, we experienced that in estimating issues, if the mathematical modeling aligns more closely with the physics of the problem at conjunction, the consequences will be more accurately estimated. Consequently, focusing on the mathematics underlying any issues is of greater significance. Furthermore, the estimation in the initial iteration loop plays a crucial role, as the subsequent loops derive their results from the same parameters. Thus, It involves accurately adjusting the main optimization function to align with the conditions of the problem.

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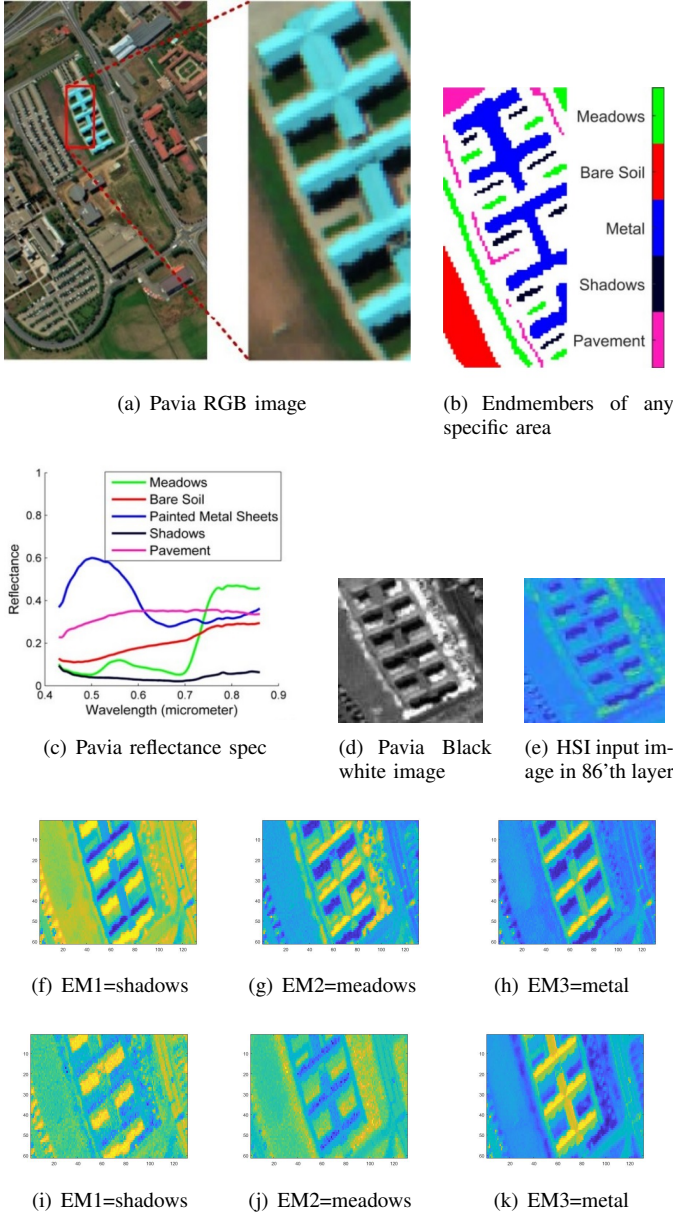


Fig. 6: Input observation data and estimated abundance maps of 3 endmembers for the second experiment with real data set; (a) Original Pavia University RGB image and the selected ROI; (b) Endmembers of any specific area; (c) EMs spectral reflectance graph for 5 EMs; (d) Black and white image of near ROI (not exact region); (e) HSI input image in 86'th layer(not exact region); (f) to (h) Estimated abundance maps based on ELMM method; (i) to (k) Estimated abundance maps by the proposed approach based on IELMM.

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